Geometrically, in the case of function f of one variable, the linear approximation at a point x_0 represents the tangent line to the graph of $y = f(x)$ at x_0. Similarly, in the case of a function F of two variables, the linear approximation at a point (x_0, y_0) represents the tangent plane to the graph of $z = F(x, y)$ at (x_0, y_0).

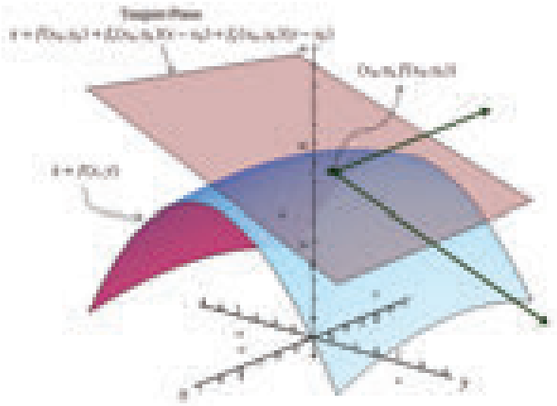


Fig. 8.13
Linear Approximation by Tangent Plane

Example 8.16

If $w(x, y, z) = x^2y + y^2z + z^2x, x, y, z \in \mathbb{R}$, find the differential dw.

Solution

First let us find w_x, w_y, and w_z.

Now $w_x = 2xy + z^2, w_y = 2yz + x^2$ and $w_z = 2zx + y^2$.

Thus,by (15), the differential is

$$dw = (2xy + z^2)dx + (2yz + x^2)dy + (2zx + y^2)dz .$$

Example 8.17

Let $U(x, y, z) = x^2 - xy + 3\sin z, x, y, z \in \mathbb{R}$. Find the linear approximation for U at $(2, -1, 0)$.

Solution

By (14), linear approximation is given by

$$L(x, y, z) = U(x_0, y_0, z_0) + \left.\frac{\partial U}{\partial x}\right|_{(x_0, y_0, z_0)} (x - x_0) + \left.\frac{\partial U}{\partial y}\right|_{(x_0, y_0, z_0)} (y - y_0) + \left.\frac{\partial U}{\partial z}\right|_{(x_0, y_0, z_0)} (z - z_0) .$$

Now $U_x = 2x - y, U_y = -x$ and $U_z = 3\cos z$.

Here $(x_0, y_0, z_0) = (2, -1, 0)$, hence $U_x(2, -1, 0) = 5$, $U_y(2, -1, 0) = -2$ and $U_z(2, -1, 0) = 3$.

Thus $L(x, y, z) = 6 + 5(x - 2) - 2(y + 1) + 3(z - 0) = 5x - 2y + 3z - 6$ is the required linear approximation for U at $(2, -1, 0)$.

EXERCISE 8.5

1. If $w(x, y) = x^3 - 3xy + 2y^2, x, y \in \mathbb{R}$, find the linear approximation for w at $(1, -1)$.

2. Let $z(x, y) = x^2y + 3xy^4, x, y \in \mathbb{R}$. Find the linear approximation for z at $(2, -1)$.

3. If $v(x, y) = x^2 - xy + \frac{1}{4}y^2 + 7, x, y \in R$, find the differential dv.

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